Performance Evaluation for Concept Learning Method from Japanese Copular Sentences

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Abstract

This paper presents a method for concept learning from Japanese copular sentences. Copular sentences typically appear in Japanese text. Our proposed method is based on heuristics by which human beings get semantic concepts from copular sentences. Our proposed method consists of the analysis of sentence structure, learning process, path reduction process, inheritance process and evaluation learning process. The system developed by this method represents the concepts of noun phrases and semantic relations between the concepts in copular sentences on concept relation network. The system represents unknown concepts with ambiguities in it. However the ambiguities are reduced as the system learns from other copular sentences. We developed an experimental system, and carried out an evaluation experiment. In this paper, we outline our method and describe the results of the evaluation experiment.

1 Introduction

It is necessary to develop a natural language understanding system in order to use computers and information more effectively. However, there are still a lot of problems with general and practical use[Tsujii 89]. Therefore we choose to research the copular sentence. The copular sentence appears in Japanese text typically. The copular sentence consists of a subject and a predicate. The subject and the predicate consist of nouns or noun phrases[Takahasi 84]. We hope that our research could be a basic technic to

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develop the system of natural language understanding. Some researchers have already presented some kinds of research on a copular sentences from the point of view of Linguistics[Takahasi 84, Kusanagi 85, Mizutani 82, Mallery 85]. However there has been little research from the point of view of Computational Linguistics. Therefore we have done research on the method of learning and analysis for concepts of noun phrases in copular sentences and the semantic relations between them[Araki 90, Araki 91, Sasaki 92, Sasaki 93].

This paper describes our proposed method on concept learning from Japanese copular sentences and the results of evaluation experiment for this method.

2 Semantic Relations of Noun Phrases

The copular sentence is "<*NP*1[1]>[2] *wa* <*NP*2> *desu*."[3]. It is translated in "<*NP*1> is <*NP*2>." Basic semantic relations between <*NP*1> and <*NP*2> are as follows:

1. **Relation of subclass concept and superclass concept**
   [*NP*2][4] is the superclass concept of [*NP*1].
   Example : *Ningen wa honyurui desu.*
   (A human is a mammal.)
2. **Relation of member concept and class concept**
   [*NP*1] is a member of [*NP*2].
   Example : *Osu wa seibetsu desu.* (Male is sex.)
3. **Relation of equivalence concept**
   [*NP*1] and [*NP*2] are equivalent.
   Example : *Fujisan wa nihon-ichi no yama desu.* (Mt. Fuji is the highest mountain of all Japanese mountains.)
4. **Relation of object concept and property concept**
   [*NP*1] is an object. [*NP*2] is a property value of [*NP*2].
   Example : *Taro wa sensei desu.* (Taro is a teacher.)
5. **Relation of object concept and event concept**

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[1]NP represents the noun phrase.
[2]<*X*> represents a lexicon of the X.
[3]An italic character represents Japanese.
[4] [*X*] represents a concept of the X.

[*NP*1] is an object. [*NP*2] is related to the event concept of [*NP*1].

Example : *Chichi ga kitaku desu.* (Father comes back home.)

6. **Relation of object concept and metaphor concept**

[*NP*2] is a metaphor of a typical property of [*NP*1].

Example : *Hanako no te wa momiji no youda.*

(Hanako's hand is like a maple)

This sentence means that Hanako's hand is small and cute.

7. **Relation of eel sentence [Okutsu 78]**

A typical example is that *Taro wa Unagi desu.* (Taro is an eel.) Normally, this sentence means that Taro eats an eel. However it is possible to interpret that Taro is the name of an eel. Therefore the interpretation of it depends on context.

8. **Relation of equivalence with same words**

[*NP*1] and [*NP*2] are same. This sentence depends on the context and is interpreted rhetorically.

Example : *Taro wa Taro desu.* (Taro is Taro.)

This sentence means that Taro should decide and do it by himself

The semantic relations from 1 to 5 are literal. The semantic relations from 6 to 8 are not literal. In this paper, our proposed method treats only the 1st, 2nd, 3rd and 4th relations. The considerations for other relations are future problems of this method.

3 Knowledge Representation and Model of Concept Generation

This method uses SNePS (Semantic Network Processing System) [Shapiro 79, Shapiro 86]. Therefore we can understand the relation of concepts intuitionally. It is easy to add new knowledge to the SNePS and modify knowledge on it. The SNePS can express a concept and a relation of concepts in the same format.

In SNePS, there is a one-to-one correspondence between nodes and represented concepts. Nodes represent intensional objects. Arcs represent structured and semantic relations. Lex arcs connect our intensional world to our external world.

In this paper, the kinds of semantic relations between concepts are limited as follows:

1. **Subclass and superclass relation**
   ← *sub* —:— *su* →[5]
   sub!su[6]
2. **Superclass and Subclass relation**
   ← *sub* —:— *sub* →
   su!sub
3. **Member and class relation**
   ← *mem* —:— *class* →
   mem!class
4. **Class and member relation**
   ← *class* —:— *mem* →
   class!mem
5. **Equivalence relation**
   ← *eq* —:— *eq* →
   eq!eq
6. **Object and property relation**
   ← *ob* —:— *prop* →
   ob!prop
7. **Property and object relation**
   ← *prop* —:— *ob* →
   prop!ob

Figure 1 shows an example of SNePS. The "Human being" and "mammal" connect to [*m*1] and [*m*3] by lex arcs respectively. The "human being" and "mammal" express lexicons. [*m*1] is the concept of "human being". [*m*3] is the concept of "mammal". [*m*2] is the concept of the semantic relationbetween [*m*1] and [*m*3] and the proposition of "*Ningen wa honyurui desu.* (A human being is a mammal.)". [*m*2] and the two arcs

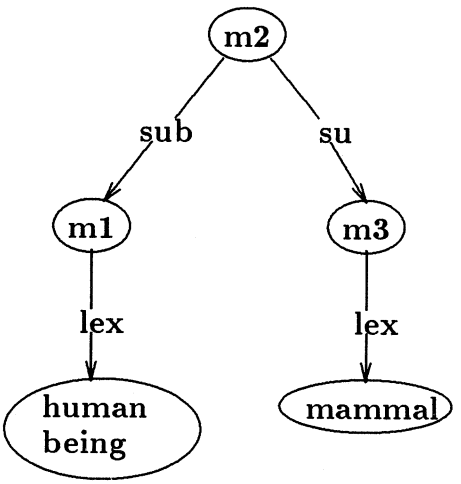


Figure 1 : Example of SNePS

[5]This representation is used on SNePS.
[6]This representation is used in the heuristics tables.

represents the subclass and superclass relation between $[m1]$ and $[m3]$.

A path is a sequence of arcs from a concept to another concept. The path has two kinds. One is a direct path that has a relation concept. The other is a lengthy path that has plural relation concepts. In our proposed method, a concept has two guidelines. One guideline is a known and unknown concept. The other guideline is a primitive and complex concept. The known concept is the concept that the SNePS has already. The unknown concept is the concept that the SNePS does not have. The lexicon of the primitive concept consists of a noun. The lexicons of the complex concept consist of plural nouns. Therefore our proposed method has four concept spaces.

A new concept is generated when an unknown concept is inputed. The new concept is generated as an unknown and primitive concept, or as an unknown and complex concept. It changes to the known concept when the process for a sentence is done. In this paper, we call a SNePS like this, the concept relation network.

4 Outlines of This Method

Figure 2 shows the outline of this method. Our proposed method decides the semantic relation between two concepts on the basis of the paths between them. The leaning process treats unknown concepts and ambiguous concepts. This method uses some heuristics in order to learn semantic concepts from copular sentences. This method expresses ambiguity on the concept relation network and reduces it by the learning process and the path reduction process when new information is inputed.

First, our proposed method analyzes the structure of the inputed copular sentence. In this process, noun phrase 1 and noun phrase 2 are extracted. The structures of them are

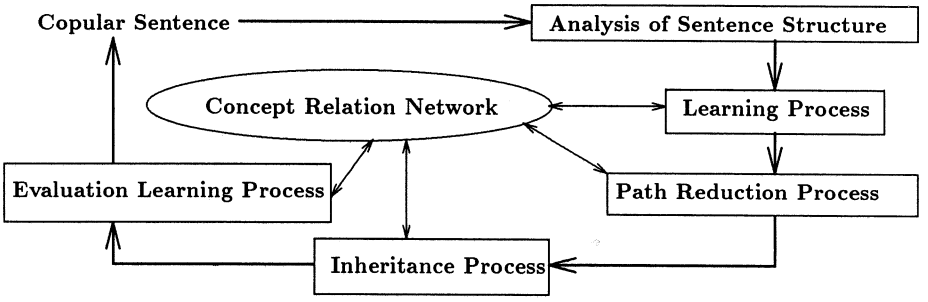


Figure 2 : Outline of Process

analyzed. Second, the learning process is performed when the copular sentence has an unkown concept or an ambiguous direct path between two concepts. Third, the path reduction process is performed. The path reduction rules of this process decide the semantic relation between two concepts on the basis of the inputed sentence and the situation of the connected concepts. The goals of this process are more effective use of knowledge and automatic adaptability of the concept relation network. Fourth, this method performs the property inheritance. Last, the evaluation learning process is performed for the recovery of erroneous semantic relations without ambiguity.

5 Concept Learning from Copular Sentences Method

5.1 Analysis of Sentence Structure

Noun phrases 1 and 2 are extracted from a copular sentence. In this method, the copular sentence is already divided into morphemes by a space. Therefore the noun phrase 1 is the morpheme between the top and "*wa, ga*". The noun phrase 2 is morpheme between "*wa, ga*" and "*desu, dearu, da* (inflectional endings)".

In this paper, the noun phrase is limited in the forms of "$<N1$[7]$>$ *no* $<N2>$" and "$<N1>$ *to*(and) $<N2>$". The "*no*" is a Japanese particle[Shimazu 86]. In "$<N1>$ *no* $<N2>$", the relations between $<N1>$ and $<N2>$ are possession, whole and part, place limitation, time limitation, material limitation, kind limitation, quantity limitation, case relations and so on. In this paper, this method treats the possession, the whole and the part, the place limitation, the kind limitation and the case relations. The inputed copular sentences include those of meta-knowledge. They are defined as sentences in which the noun phrase 2 is the lexicon for the definition of the concept relation.

5.2 Learning Process

This process generates unkown concepts and decides semantic relations between concepts by some heuristics. These heuristics have been created comprehensively. The learning heuristics have two kinds. One depends on the structure of the inputed sentence. The other depends on the situation of the connected concepts. The former has seven kinds. Table 1 shows them. Those are expressed by LHSS (Learning Heuristics depend-

[7] N represens the noun.

ed on Structure of Sentence). The later has three kinds. Tables 2, 3 and 4 show them. Those are expressed by LHSC (Learning Heuristics depended on Situation of the connected Concepts).

LHSC is based on the situation of the connected concepts. Figure 3 shows their situation. In this figure, A, B, C1 and C2 are nodes that represent concepts. R, R1 and C2

Table 1 : Learning Heuristics based on the Structure of the Sentence

Class	No.	Object	Relations
1		<*NP*1>*wa*<*NP*2>*desu*. (<*NP*1>is<*NP*2>)	
	1	<*NP*1>:<*NP*2>	sub!su mem!class eq ob!prop
2		<*NP*>is″<*N*1>*no*<*N*2>.″	
	2	<*N*1>:<*N*2>	ob!prop
	3	<*N*1>*no*<*N*2>:<*N*1>	ob!prop
	4	<*N*1>*no*<*N*2>:<*N*2>	sub!su mem!class
3		<*NP*>is″<*N*1>*to*<*N*2>.″	
	5	<*N*1>:<*N*1>*no*<*N*2>	sub!su mem!class
	6	<*N*2>:<*N*1>*no*<*N*2>	sub!su mem!class
4		<*N*1>*no*<*N*2>*wa*<*N*3>*desu*. (<*N*1>*no*<*N*2>*is*<*N*3>.)	
	7	<*N*1>*no*<*N*2>:<*N*3>	eq
	8	<*N*2>:<*N*3>	su!sub class!mem
5		<*N*1>*to*<*N*2>*wa*<*N*3>*desu*. (<*N*1>*and*<*N*2>*is*<*N*3>.)	
	9	<*N1*>*to*<*N*2>:<*N*3>	sub!su eq
	10	<*N*1>:<*N*3>	sub!su mem!class
	11	<*N*2>:<*N*3>	sub!su mem!class
6		<*N*1>*wa*<*N*2>*no*<*N*3>*desu*. (<*N*1>*is*<*N*2>*no*<*N*3>.)	
	12	<*N*1>:<*N*2>*no*<*N*3>	sub!su mem!class eq
	13	<*N*1>:<*N*3>	sub!su mem!class
7		<*N*1>*wa*<*N*2>*to*<*N*3>*desu*. (<*N*1>*is*<*N*2>*and*<*N*3>.)	
	14	<*N*1>:<*N*2>*to*<*N*3>	su!sub class!mem
	15	<*N*1>:<*N*2>	su!sub class!mem
	16	<*N*1>:<*N*3>	su!sub class!mem

Table 2 : LHSC : R based on R1

R1	sub!su	mem!class	su!sub	class!mem	eq	ob!prop	prop!ob
R	–	eq prop!ob	–	–	–	sub!su mem!class eq	–

"–" represents that R is not limited.

Table 3 : LHSC : R based on R2

R2	sub!su	mem!class	su!sub	class!mem	eq	ob!prop	prop!ob
R	sub!su eq ob!prop	mem!class eq ob!prop	sub!su eq ob!prop	sub!su eq ob!prop	–	–	sub!su eq

are arcs that are semantic relations between concepts. Table 2 shows the heuristics for R based on R1. Table 3 shows the heuristics for R based on R2. Table 4 shows the heuristics for R based on R1 and R2.

Table 4 : LHSC : R based on R1 and R2

R2	R1						
	sub!su	mem!class	su!sub	class!mem	eq	ob!prop	prop!ob
sub!su	sub!su eq ob!mem	X1	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq	sub!su eq ob!prop
mem!class	mem!class ob!prop	eq	mem!class ob!prop	mem!class ob!prop	mem!class eq ob!prop	mem!class eq	ob!prop
su!sub	sub!su eq ob!prop	X1	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq	sub!su eq ob!prop
class!mem	sub!su eq ob!prop	X1	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq ob!prop	sub!su eq	sub!su eq ob!prop
eq	—	X2	—	—	—	sub!su mem!class ep	—
ob!prop	—	X2	—	—	—	sub!su mem!class eq	—
prop!ob	sub!su eq	X1	sub!su eq	sub!su eq	sub!su eq	X3	sub!su eq

X1 represents no relation or "class!mem". X2 represents no relation or "eq". X3 represents no relation or "prop!ob".

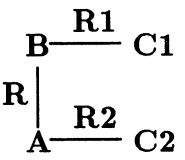


Figure 3 : Situation of the Connected Concepts

5.3 Certainty Degree

Our proposed method uses certainty degree for the relations between concepts. This degree is expressed by CD (Certainty Degree). The values of CD are from 0.01 to 1.00. The value 1.00 is the most certain. The value 0.01 is the most ambiguous. All nodes except lexical nodes have CD's. The formula of the calculation of CD is as follows:

$$CD = \frac{\sum_{j=1}^{N} CD_j}{N} \quad (1)$$

when N is the total number of nodes between two concepts.

All the CD's for a direct path and all lengthy paths between two concepts are calcu-

lated by the formula (1). The highest value is decided as the CD of the node that expresses the semantic relation between the two concepts.

5.4 Restriction of Searched Space

Human beings do not recall all the concepts connected with a concept when they try to recall the concept. They forget concepts that are not impressive. Some concepts are too impressive to recall the connected concepts. Therefore our proposed method uses active degree for a situation like this. It is expressed as AD (Active Degree). AD is defined as follows:

$$AD=\alpha\times FU-\beta\times FN+\gamma\times CD \tag{2}$$

when α, β, γ are coefficients, the FU is frequency for use and the FN is frequency for no use.

The searched space in this system is restricted by AD. The conditions of limitation of the searched space are as follows:

1. The AD's for all nodes are over 0.
2. The difference between a node and connected node is less than fifty.

5.5 Path Reduction Process

Our proposed method uses the learning heuristics. However the ambiguity of semantic relations still remains in the concept relation network. Therefore the path reduction process is introduced to this method for more effective reduction of the ambiguity[Sasaki 93]. Path reduction rules are used in the path reduction process. They are automatically generated on the basis of the situation of the connected concepts. This process gives not only the reduction of ambiguity but also gives generality to this method.

The left-hand side of the path reduction rule is the specific lengthy path and the right-hand side is the direct path without ambiguity. The conditions for extraction of the specific lengthy path are as follows:

1. $<NP1>$ is the first node and $<NP2>$ is the terminal node.
2. It is the shortest among all the lengthy paths.
3. All relation concepts on the lengthy path have no ambiguity or only one of them is ambiguous.

An example of the path reduction rule is as follows:

$$[mX] \leftarrow sub!su \rightarrow [mY] \leftarrow sub!su,\ mem!class \rightarrow [mZ]$$
$$\Downarrow$$
$$[mX] \leftarrow sub!su \rightarrow [mZ]$$

5.6 Inheritance Process

The property of the concept of "$<N1>$ *no* $<N2>$" depends on that of $[N1]$. Therefore, we consider that the property inheritance from the concept of "$<N1>$ *no* <N2>" to $[N1]$ is possible[Fraser 92, Finin 84]. The inheritance process of this method treats only "$<N1>$ *no* $<N2>$". New $[\varepsilon]$ is generated when the situation of the concept relation network satisfies the inheritance heuristics[Sasaki 93] as follows:

1. Conditions of relations between concepts
   (a) The relation between a $[\alpha]$ and a $[\beta]$ is sub!su or mem!class.
   (b) The relation between the $[\beta]$ and a $[\gamma]$ is ob!prop.
   (c) The relation between the $[\gamma]$ and a $[\delta]$ is sub!su.
   (d) The relation between the $[\alpha]$ and the $[\epsilon]$ is ob!prop.
   (e) The relation between the $[\varepsilon]$ and the $[\gamma]$ is sub!su.
2. Conditions of structures of lexicons of concepts
   (a) The noun phrase structure of lexicon of the $[\gamma]$ is "$<\beta>$ no $<\delta>$"
   (b) The noun phrase structure of lexicon of the $[\varepsilon]$ is "$<\alpha>$ no $<\delta>$".

5.7 Evaluation Learning Process

This process recovers an erroneous semantic relation that has already been decided without ambiguity. The errors occurred by a change of situation and erroneous applications of the heuristics. This process has some heuristics[Araki 91].

6 Evaluation Experiment

6.1 Data and Procedure

Fifty copular sentences are inputed to the experimental system developed by our method. The first five sentences of them are as follows:

1. *Zo wa honyurui desu.*(An elephant is a mammal.)
2. *Ningen to zo wa doubutsu desu.*(A human being and an elephant are animals.)
3. *Honyurui wa doubutsu desu.*(A mammal is an animal.)

4. *Ningen wa doubutsu desu.*(A human being is an animal.)

5. *Ningen to zo wa honyurui desu.*(A human being and an elephant are mammals.)

Figure 4 shows the initial world for this experiment.

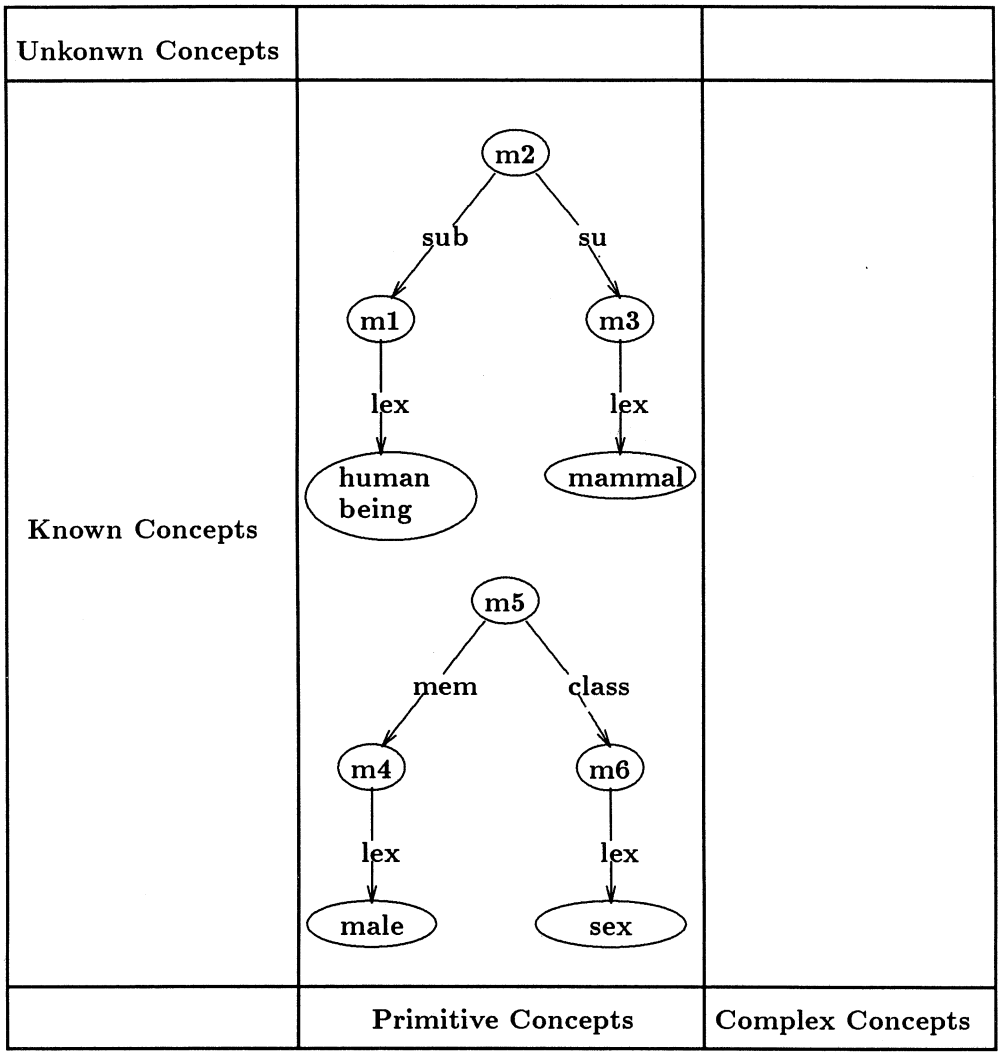


Figure 4 : Initial World

6.2 Results and Considerations

Figure 5 shows the world after learning from the first sentence. The heuristics applied for the first sentence are LHSS1 of Table 1. Figure 6 shows the world after learning two sentences. The heuristics applied for the second sentence are LHSS1, 5, 6, 9, 10 and 11 of Table 1.

The path reduction rules were generated in seven sentences and applied in six sentences. The inheritance process performs in two sentences. Figure 7 shows the change of ambiguity of the concept relation network. Ambiguity degree is expressed by AMD (AMbiguity Degree). AMD is defined as follows:

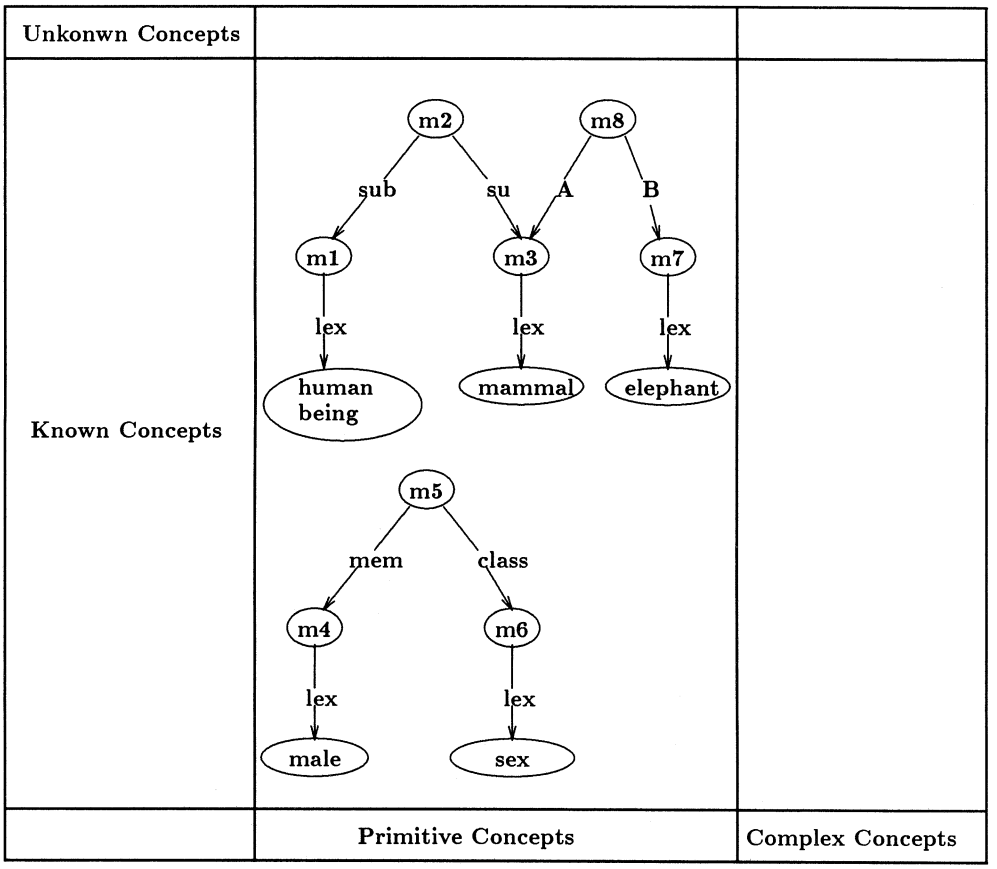


Figure 5 : World after Learning from the First Sentence

$$AMD=\frac{\text{Total number of relations}}{\text{Total number of arcs}} \tag{3}$$

The AMD is from 1.3 to 1.4 after learning from fifteen sentences. This result shows that this method is effective for reduction of ambiguity.

Figure 8 shows the world after learning from fifty sentences. In this world, all

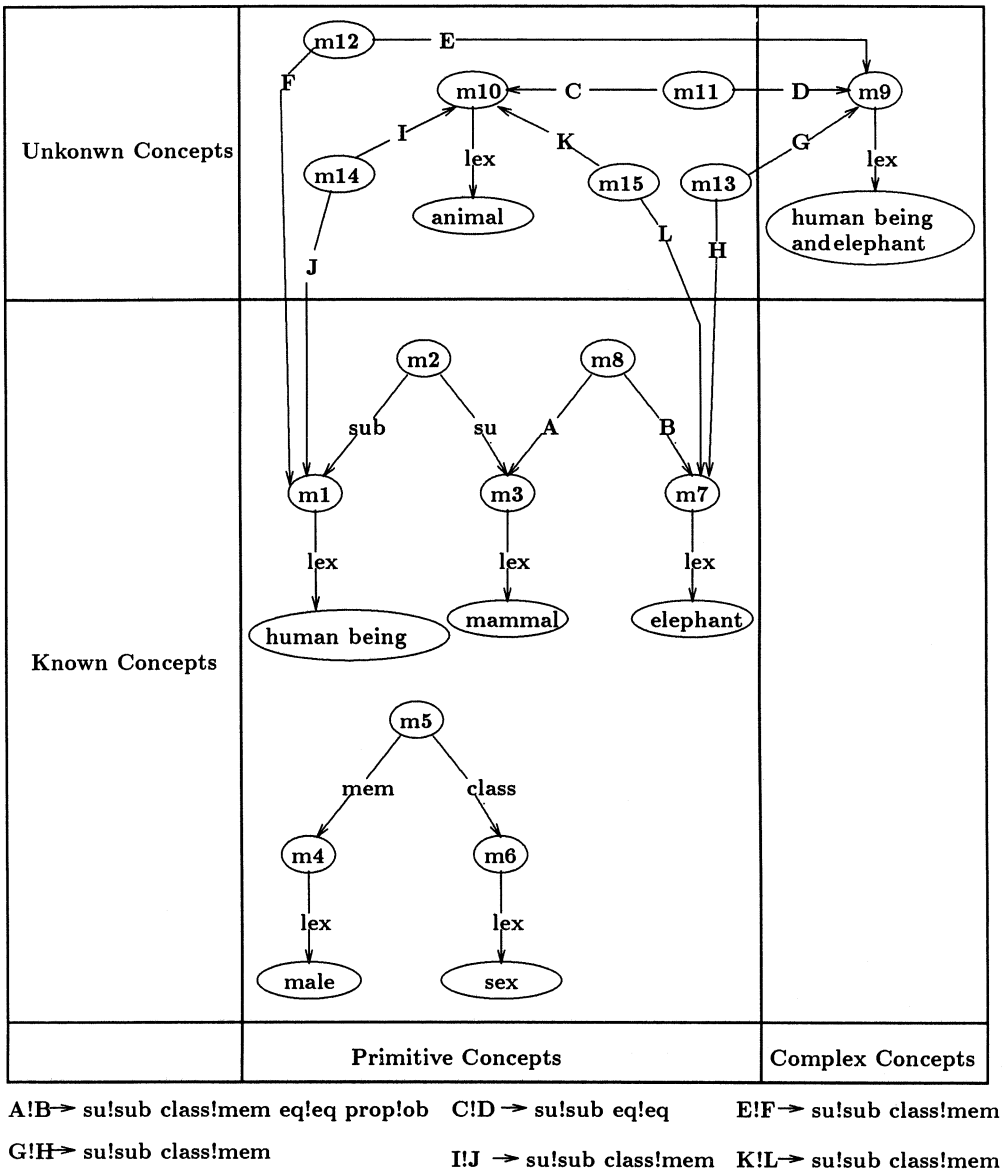


Figure 6 : World after Learning from Two Sentences

relations decided without ambiguity are correct and all relations with ambiguity include a correct relation respectively.

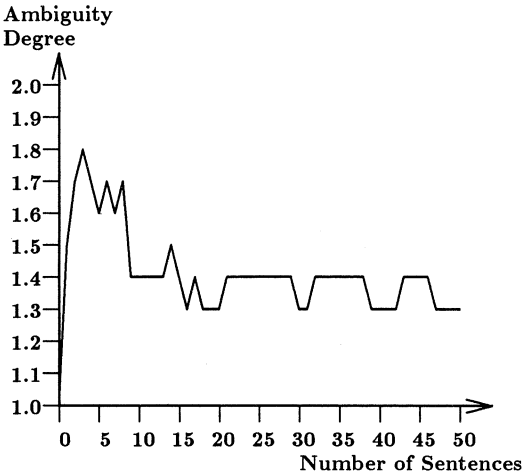


Figure 7 : Change of Ambiguity

7 Conclusions

In this paper, we proposed the method for learning concepts and relations between the concepts found in copular sentences. This method gives a basis for realization of natural language understanding system. Our proposed method is especially effective for the reduction of ambiguity. This method uses some heuristics and the path reduction rules. We developed the experimental system on the basis of this method and have done the evaluation experiment. Our result was that the ambiguity degree was about 1.3 after learning from fifty sentences. All relations are correct or include a correct relation respectively in the world after learning from fifty sentences. This results shows that this method is effective as the basis of developing a natural language understanding system.

Future problems for this method include the extension for other structures of a noun phrase and the extension of other parts of speech. The goal of this research is that this method can learn from standard sentences by using this approach.

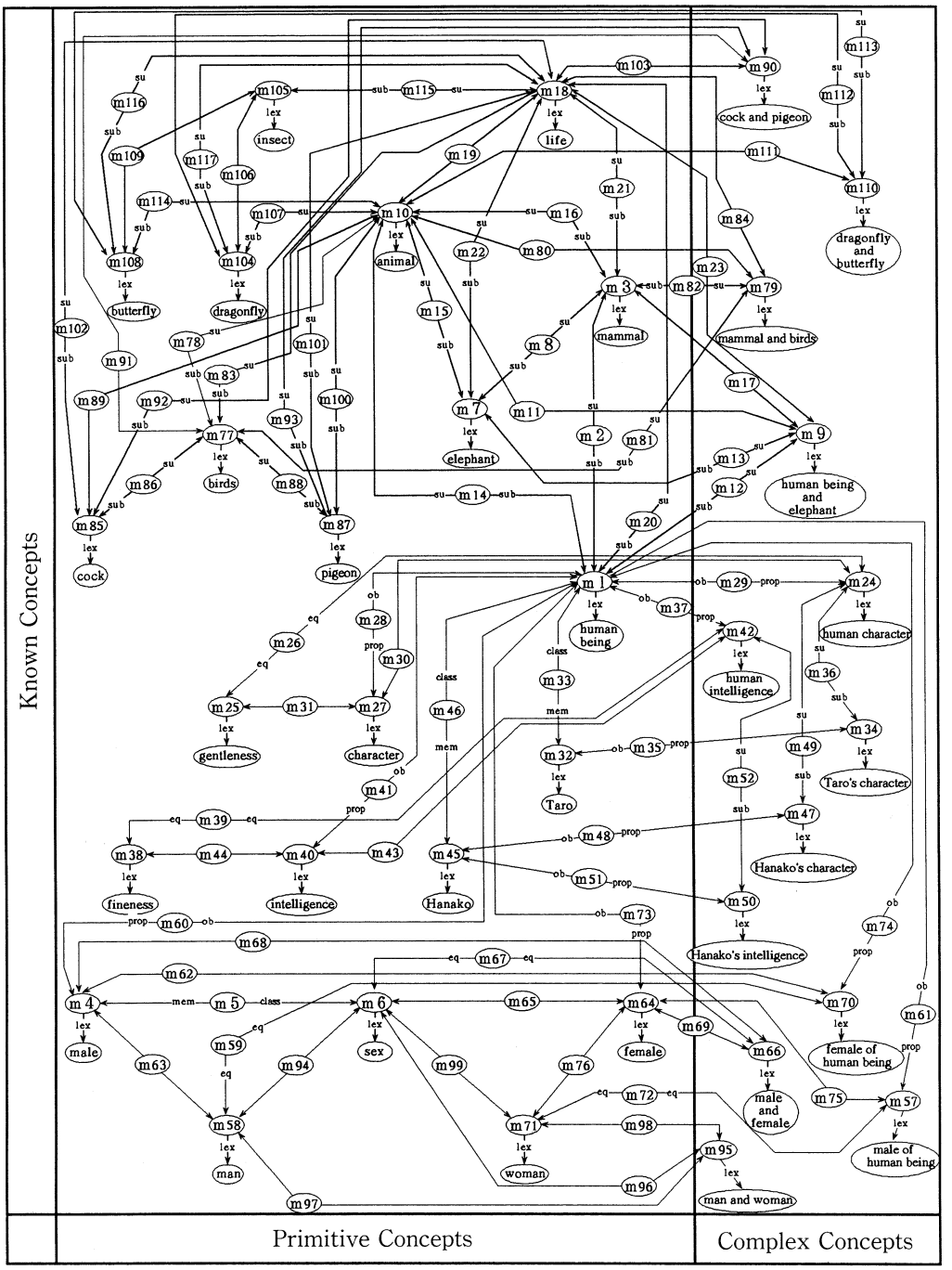


Figure 8 : World after Learning from Fifty Sentences

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